Ex. 2.2–2 Diet Problem

Ozark farms uses at least 800lb of special feed daily (mixture of corn and soybean)

Feedstuff	Protein	Fiber	Cost ($/lb)
Corn	0.09	0.02	0.30
Soybean	0.60	0.06	0.90

Dietary requirements: → at least 30% protein ✓
at most 5% fiber ✓

The amount of protein included in x_1 lb of corn and x_2 lb of soybean meal is
$0.09x_1 + 0.60x_2$
This quantity should equal at least 30% of the total feed mix (x_1+x_2) lb.
That is → $0.09x_1 + 0.60x_2 \geq 0.30(x_1+x_2)$
Similarly $0.02x_1 + 0.06x_2 \leq 0.05(x_1+x_2)$

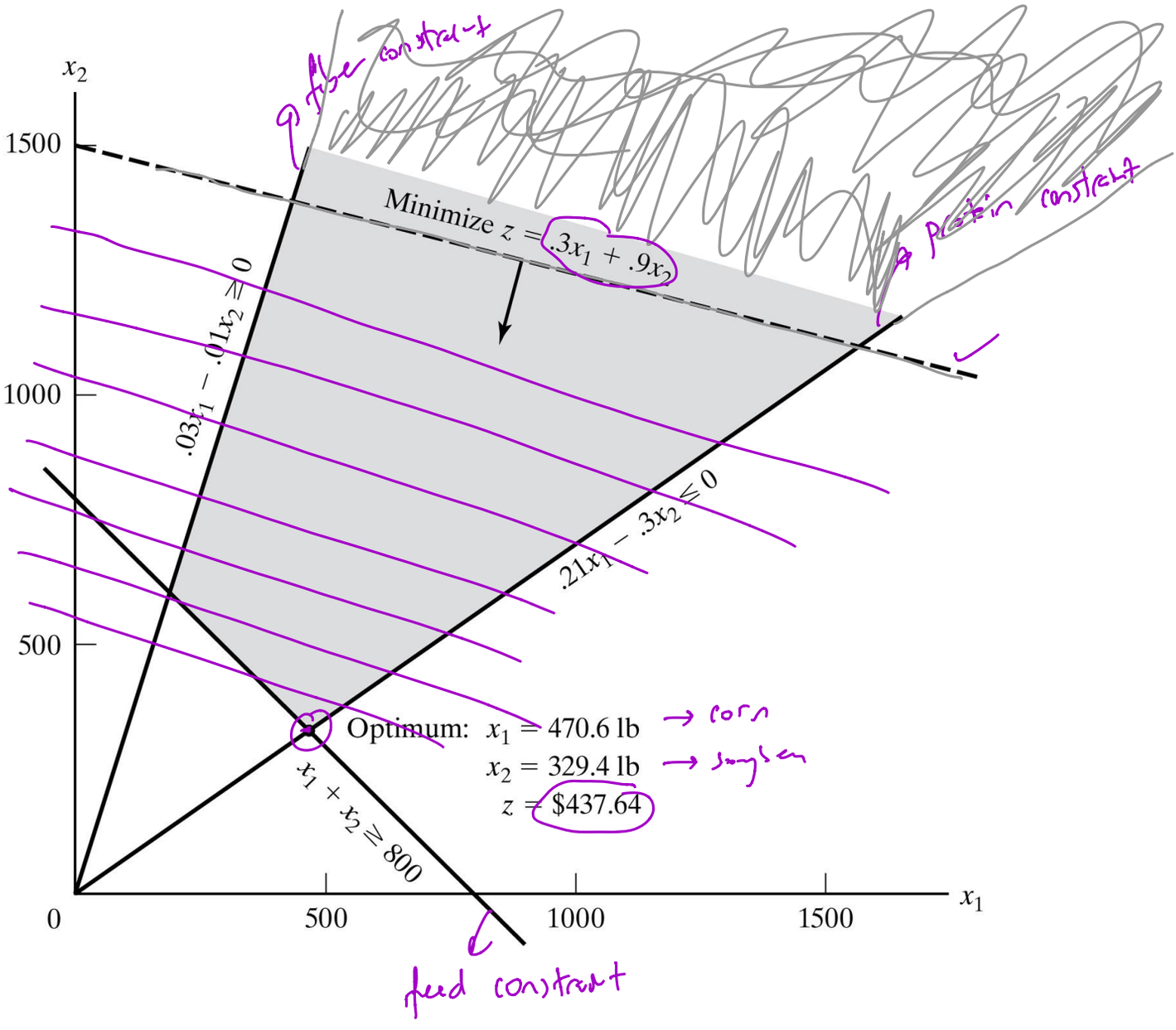
- What is the minimum cost feed mix?

x_1 = amount of corn in the mix
x_2 = " soybean "

minimize $z = 0.3x_1 + 0.9x_2$

s.t.
$x_1 + x_2 \geq 800$
$0.09x_1 + 0.60x_2 \geq 0.3(x_1+x_2)$
$0.02x_1 + 0.06x_2 \leq 0.05(x_1+x_2)$
$x_1, x_2 \geq 0$

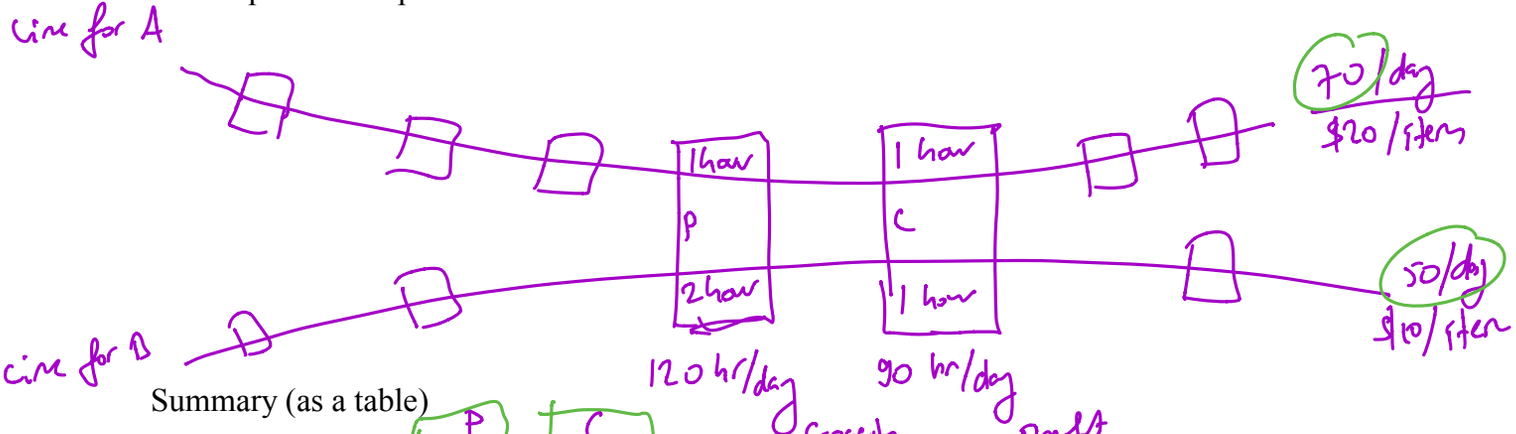
Min. $z = 0.3x_1 + 0.9x_2$
s.t.
$x_1 + x_2 \geq 800$
$0.21x_1 - 0.3x_2 \leq 0$
$0.03x_1 - 0.01x_2 \geq 0$
$x_1, x_2 \geq 0$


x_2
1500
1000
500
0
500
1000
1500
x_1
Minimize $z = .3x_1 + .9x_2$
$.03x_1 - .01x_2 \geq 0$
$.21x_1 - .3x_2 \leq 0$
$x_1 + x_2 \geq 800$
Optimum: $x_1 = 470.6$ lb
$x_2 = 329.4$ lb
$z = \$437.64$

LP MODELING EXAMPLES

A Product Mix Problem:

- A TV company produces two types of models, A and B.
- There are two production lines, one for each model.
  - The capacity of the A production line is 70 sets per day,
  - The capacity of the B line is 50 sets per day.
- There are two departments P (picture tube) and C (chassis) both of which are used in the production of each model.
  - In department P, model A requires 1 labor hour and B requires 2 labor hours; a maximum of 120 labor hours can be assigned to department P.
  - In C, A requires 1 hour and B requires 1 hour; a maximum of 90 labor hours can be assigned to C.
- Profit contributions are $20 and $10 per set of A and B, respectively.
- If the company can sell as many sets as it can produce, what should be the daily production plan for each model?



Summary (as a table)

	P	C	Capacity	Profit
A	1	1	70	20
B	2	1	50	10
Capacity →	120	90		

Decision variables:

x_A : daily production of A

x_B : " " of B

Max. $z = 20 x_A + 10 x_B$

s.t.

$x_A + 2x_B \le 120$ (P's capacity)

$x_A + x_B \le 90$ (C's capacity)

$x_A \le 70$ (A's capacity)

$x_B \le 50$ (B's capacity)

$x_A, x_B \in \mathbb{Z}^+$

$x_A, x_B \ge 0$

Important note: to be correct, the variables should be set to non-negative integers. The model then would be ILP, not LP.

A Blending Problem: The Green Lawn & Garden store is trying to sell its own brand of lawn fertilizer this year. It plans to sell two types of fertilizer, one high in nitrogen content and the other an all-purpose fertilizer. The fertilizers are mixed from two ingredients that contribute nitrogen and phosphorus in different amounts. The composition of each ingredient and cost per pound is given below.

Ingredient	Nitrogen, %	Phosphorus, %	Cost, $/lb
1	60	10	0.20
2	10	40	0.30

The season's demand is estimated to be 5,000 25-lb bags of high-nitrogen fertilizer and 7,000 25-lb bags of all-purpose fertilizer. The high-nitrogen fertilizer is to contain between 40 to 50% nitrogen and all-purpose fertilizer is to contain, at most, 20% phosphorus. How many pounds of each ingredient should the store purchase to satisfy the estimated demand at minimum cost?

Decision variables:

x_{ij} = amount of ingredient i purchased for fertilizer j → (i: 1 = HN, 2 = AP)

Objective Function:

Minimize $z = 0.20(x_{11}+x_{12}) + 0.30(x_{21}+x_{22})$

$= 0.2x_{11} + 0.2x_{12} + 0.3x_{21} + 0.2x_{22}$

Constraints:

$x_{11} + x_{21} \geq 125000$ → demand of HN

$x_{12} + x_{22} \geq 175000$ → " AP

$0.60x_{11} + 0.10x_{21} \geq 0.40(x_{11}+x_{21})$ → minimum nitro content in HN

$0.60x_{11} + 0.10x_{21} \leq 0.50(x_{11}+x_{21})$ → maximum " " " HN

$0.10x_{12} + 0.40x_{22} \leq 0.20(x_{12}+x_{22})$ → maximum phosp. " " AP

$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$

The complete Model:

minimize $z = 0.20x_{11} + 0.20x_{12} + 0.30x_{21} + 0.20x_{22}$

$$x_{11} + x_{21} \geq 125000$$

$$x_{12} + x_{22} \geq 175000$$

$$0.20x_{11} - 0.30x_{21} \geq 0$$

$$0.10x_{11} - 0.40x_{21} \leq 0$$

$$-0.10x_{12} + 0.20x_{22} \leq 0$$

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$$

The Complete LP-Model:

A Scheduling Problem: A personnel manager must schedule the security force in such a way as to satisfy the staffing requirements shown below:

Work time interval	1	2	3	4	5	6
Minimum # required	5	7	15	7	12	9

Staffing requirements time intervals are,

midnight-4am, 4am-8am, 8am-noon, noon-4pm, 4pm-8pm, and 8pm-midnight.

Officers work 8-hr shifts:

midnight-8am, 4am-noon, 8am-4pm, noon-8pm, 4pm-midnight, and 8pm-4am.

The manager wants to know how many officers should work each shift in order to minimize the total number of officers employed.

Decision variable:

x_i : number of officers start in work shift i.

Shift	midnight to 4am	4am to 8am	8am to noon	noon to 4pm	4pm to 8pm	8pm to midnight
1	x_1	x_1				
2		$+ x_2$	x_2			
3			$+ x_3$	x_3		
4				$+ x_4$	x_4	
5					$+ x_5$	x_5
6	$+ x_6$					$+ x_6$
	≥ 5	≥ 7	≥ 15	≥ 7	≥ 12	≥ 9

Mathematical Model

$$\text{minimize } z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

$$x_1 + x_6 \geq 5$$

$$x_1 + x_2 \geq 7$$

$$x_2 + x_3 \geq 15$$

$$x_3 + x_4 \geq 7$$

$$x_4 + x_5 \geq 12$$

$$x_5 + x_6 \geq 9$$

$$x_i \geq 0, \quad i = 1, \ldots, 6$$